

Mathematics for Machine Learning

Part I: Functions, Calculus and Optimization

Sourav Karmakar

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- 4 Multivariable Calculus & Optimization
- 5 Wrapping Up

How to Read This Deck

A Note for Engineers Learning ML

Our Guiding Principle

Every formula in this deck exists to answer one practical question:

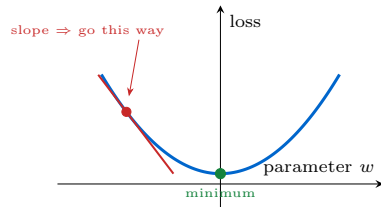
“How do I make a model’s error smaller?”

Wherever possible we lead with the **picture**, then the formula.

Two Ideas That Carry Everything

- 1 A **derivative** is the **slope** of a curve — it tells you which way is downhill.
- 2 An **integral** is the **area** under a curve — it tells you how much accumulated.

If you remember nothing else, remember these two.



Training a model = walking downhill on this curve.

Mappings, Functions & Their Shapes

Start Here: A Mapping Between Two Sets

Before we say the word “function”

The Basic Picture

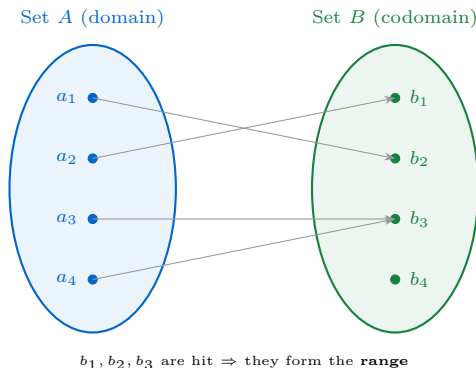
A **mapping** takes each item in a set A and points it at an item in a set B .

- A is the **domain** — what goes in.
- B is the **codomain** — where arrows may land.
- The items actually hit form the **range**.

The One Rule

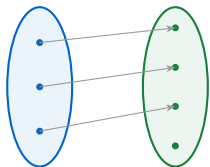
Every item in A sends out **exactly one** arrow. Never zero, never two.

If $x_1 = x_2$ then $f(x_1) = f(x_2)$.



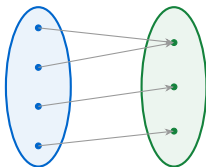
Four Kinds of Mapping

The Vocabulary, Seen All At Once



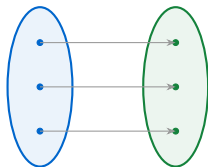
One-to-one
(injective)

distinct in
 $\Rightarrow$ distinct out



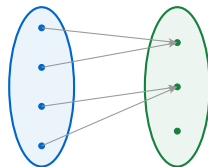
Onto
(surjective)

nothing in B
is left unhit



Bijective
(both)

invertible!
perfect pairing



Many-to-one

information lost
cannot invert

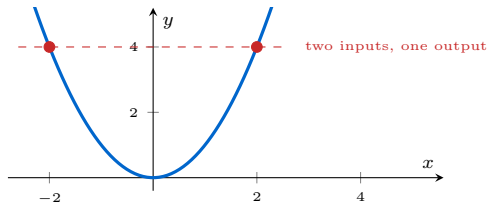
Why an Engineer Should Care: “Can I Undo It?”

Invertibility is the Only Thing You Must Remember

The Inverse Function

The inverse f^{-1} exists **only when f is a bijection**.

- $y = e^x$ is a bijection onto $(0, \infty) \Rightarrow \ln$ undoes it.
- $y = x^2$ on all of $\mathbb{R}$ is many-to-one:
 $f(-2) = f(2) = 4$.
- Given only $y = 4$, you **cannot** recover the sign of x .



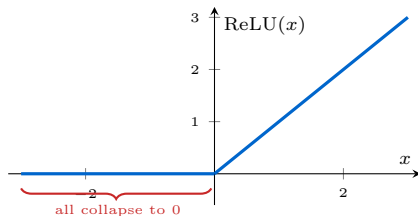
Horizontal line test: if any horizontal line cuts the curve twice, the function is not invertible.

Where Invertibility Bites in Machine Learning

Three Places You Will Meet This Again

Information Is Destroyed On Purpose

- **ReLU** maps every negative input to 0. Strictly many-to-one — once a value is zeroed, the network cannot tell -3 from -70 , and the gradient signal for those inputs vanishes.
- **Loss functions** are many-to-one by design: many different parameter settings give the same error. This is why more than one “good” model exists.



Information Is Preserved On Purpose

- **Normalizing flows** are assembled *only* out of bijections, precisely so the model can be run backwards to generate samples.

Everything on the left half is mapped to the same value — that half of the input is unrecoverable.

Linear Functions

The Straight Line and Its Slope

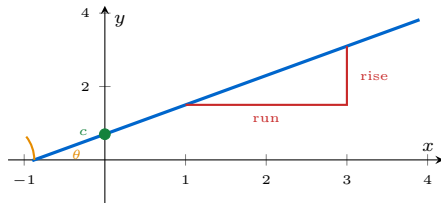
From General Form to Slope-Intercept

A straight line in general form uses d for the constant term, so that c is free to mean the intercept:

$$ax + by + d = 0$$

Solving for y :

$$y = \underbrace{-\frac{a}{b}}_m x - \underbrace{\frac{d}{b}}_c \implies \boxed{y = mx + c}$$



$$m = \frac{\text{rise}}{\text{run}} = \tan \theta$$

Reading the Two Numbers

$m = \text{slope} = \tan \theta = \text{rise/run}$ (*how steep*).

$c = \text{y-intercept}$ (*where it crosses the y-axis*).

Linear Functions: Worked Example

Extracting Slope and Intercept

Analyse $2x + 5y - 7 = 0$

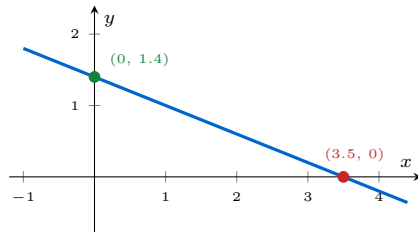
Solve for y :

$$5y = -2x + 7$$

$$y = \left(-\frac{2}{5}\right)x + \frac{7}{5}$$

Comparing with $y = mx + c$:

- **Slope** $m = -\frac{2}{5} = -0.4$
(negative $\Rightarrow$ line falls left-to-right)
- **Intercept** $c = \frac{7}{5} = 1.4$



The ML Connection

Linear regression is exactly this line — the whole job of “training” is choosing the best m and c .

Polynomial Functions

Adding Degrees Adds Bends

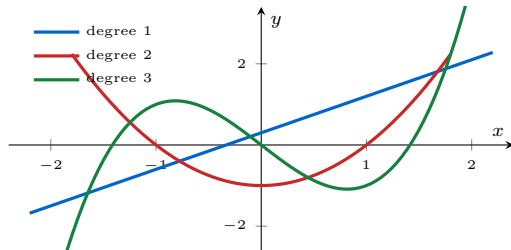
General Form

$$y = a_0 + a_1x + a_2x^2 + \cdots + a_nx^n$$

- Degree 1 — straight line, no bends
- Degree 2 — parabola, one bend
- Degree 3 — cubic, up to two bends

Rule of Thumb

A degree- n polynomial has **at most** $n - 1$ **turning points**. More degree = more wiggle = more capacity to **overfit**.



Exponential Functions

Growth and Decay

The Form

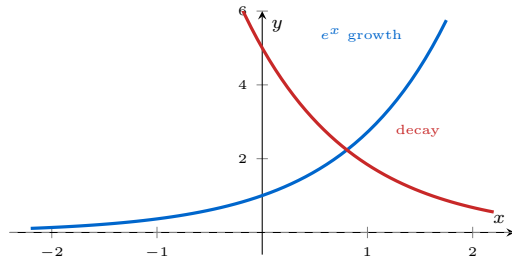
$$y = c \cdot a^{bx} \quad \text{or} \quad y = e^x$$

with Euler's number $e \approx 2.71828$.

- $b > 0 \Rightarrow$ **growth** (explodes)
- $b < 0 \Rightarrow$ **decay** (dies to 0)
- Always stays **positive**

Seen In ML As

- Learning-rate decay schedules
- The e^{-u} inside the sigmoid
- e^{z_i} inside softmax



Decay curves flatten toward zero but never touch it.

Logarithms: The Undo Button for Exponentials

Mirror Image About $y = x$

Definition

$$y = \log_a(x) \iff x = a^y$$

valid for $x > 0$, $a > 0$, $a \neq 1$.

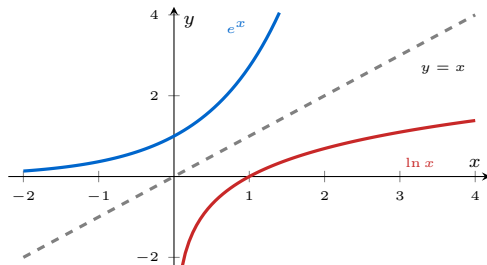
Because e^x is a **bijection**, the log exists — this is the invertibility idea from earlier, cashed in.

The Three Properties That Matter

$$\log_a(x^p) = p \log_a(x)$$

$$\log_a(x \cdot y) = \log_a(x) + \log_a(y)$$

$$\log\left(\prod_i x_i\right) = \sum_i \log(x_i)$$



Each curve is the other reflected in the dashed line.

The Killer Application: Numerical Underflow

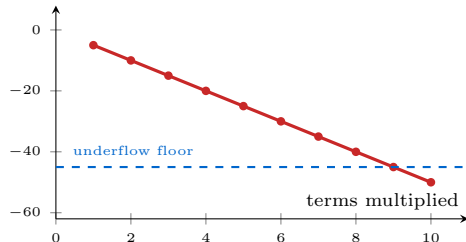
Why Every ML Library Works in Log-Space

The Problem

Multiply 10 independent probabilities, each $p_i \approx 10^{-5}$ (Naive Bayes, likelihoods):

$$\prod_{i=1}^{10} p_i \approx 10^{-50}$$

Push on to 100 terms and you reach 10^{-500} — below the smallest number the hardware can represent. It is silently **flushed to 0**, and the model now reports probability zero for everything.



Below the dashed line, the hardware simply stores 0.

The Fix: Work With Log-Probabilities

Products Become Sums

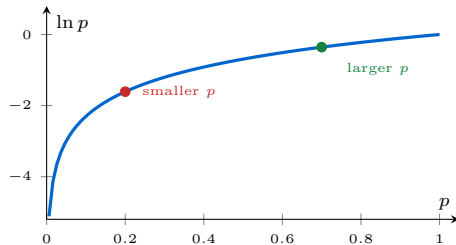
Take Logarithms

$$\log \left(\prod_{i=1}^n p_i \right) = \sum_{i=1}^n \log(p_i)$$

For our example: $\sum_{i=1}^{10} \log_{10}(p_i) = -50$, an utterly ordinary number. **No underflow.**

Why This Is Safe

$\log$ is **strictly increasing**: the largest p still has the largest $\log p$, so ranking and maximising are unaffected.



Order is preserved: larger p always gives larger $\ln p$.
Ranking is unaffected.

Section Take-Away: Mappings & Functions

- ① A **function** is a mapping where every input sends out exactly one arrow.
- ② The only classification that really matters day-to-day: **can I undo it?** Bijection $\Rightarrow$ yes. Many-to-one $\Rightarrow$ no, information is permanently lost.
- ③ **Slope** m answers “how steep”, **intercept** c answers “where does it start”. Linear regression is nothing more than picking these two.
- ④ **Higher polynomial degree = more wiggle = more overfitting risk.**
- ⑤ **Logs turn products into sums.** This single property is what keeps probability calculations from collapsing to zero on real hardware.

Next: if a curve bends, what is its slope? $\Rightarrow$ **Differentiation**

Chapter 1 Quiz

Mappings, Functions & Their Shapes

- 1 Is $f(x) = x^2$ invertible over all of $\mathbb{R}$? Explain in one line.
- 2 A line has equation $3x + 4y - 8 = 0$. Read off its slope and its y -intercept.
- 3 Why does taking logarithms prevent numerical underflow?
- 4 ReLU sends both -3 and -70 to 0 . What kind of mapping is this, and what is lost?
- 5 A degree-5 polynomial has at most how many turning points? What is the ML risk of a high degree?
- 6 True or false: $\log(a \cdot b) = \log a + \log b$.

Answers are in the **Appendix** at the end of the deck.

Differentiation: The Slope of a Curve

The Core Idea, In One Picture

A Derivative Is Just a Slope

For a **straight line** the slope is the same everywhere: rise over run.

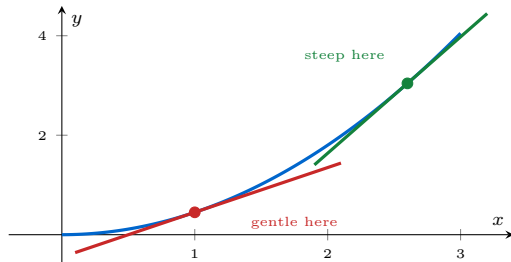
Curves Are the Problem

On a curve the steepness **changes at every point**. So we ask a narrower question:

*“How steep is the curve **right here**?”*

The answer is the slope of the **tangent line** that just grazes the curve at that point.

$\frac{dy}{dx}$ at a point = slope of the tangent there.



Same curve, two points, two very different slopes.

From Secant to Tangent

Where the Limit Comes From

Step 1 — Use Two Points

Pick x and a nearby $x + \Delta x$. The line joining them (a **secant**) has an easy slope:

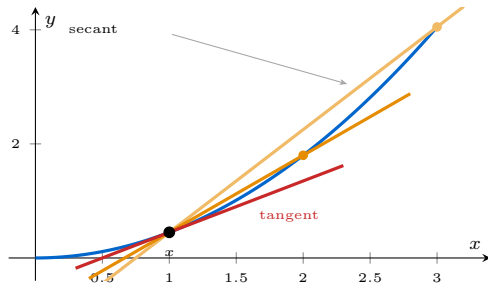
$$\frac{\Delta y}{\Delta x} = \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

Step 2 — Slide Them Together

Shrink Δx toward 0. The secant pivots until it becomes the **tangent**:

$$\frac{dy}{dx} = f'(x) = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x) - f(x)}{\Delta x}$$

In words: the limit of “rise over run” as the run shrinks to nothing.



As the second point slides in, the secant rotates onto the tangent.

When the Slope Does Not Exist

Corners Break Differentiability

The Condition

A function is differentiable at x only if approaching from the left and from the right give the **same** slope:

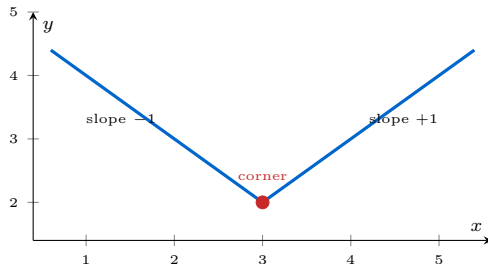
$$\lim_{\Delta x \rightarrow 0^-} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \rightarrow 0^+} \frac{\Delta y}{\Delta x}$$

Counter-Example: $y = 2 + |x - 3|$

At the corner $x = 3$:

- from the left the slope is -1
- from the right the slope is $+1$

They disagree, so $\frac{dy}{dx}$ is **undefined** at $x = 3$.



ML note: ReLU has exactly this corner at 0. Libraries simply *pick* a value (usually 0) — the “subgradient” — and training works fine in practice.

The Rules of Differentiation

Your Working Toolkit

The Three Structural Rules

Rule	Formula	Example
Linearity	$\frac{d}{dx}[\alpha f + \beta g] = \alpha f' + \beta g'$	$\frac{d}{dx}[2x^2 + 3e^x] = 4x + 3e^x$
Product	$\frac{d}{dx}[fg] = fg' + gf'$	$\frac{d}{dx}[e^x \sin x] = e^x(\sin x + \cos x)$
Quotient	$\frac{d}{dx}\left[\frac{f}{g}\right] = \frac{gf' - fg'}{g^2}$	$\frac{d}{dx}[\tan x] = \sec^2 x$

Standard Derivatives Worth Memorising

$$\frac{d}{dx}(x^n) = nx^{n-1} \quad \frac{d}{dx}(e^x) = e^x \quad \frac{d}{dx}(\ln x) = \frac{1}{x} \quad \frac{d}{dx}(\sin x) = \cos x \quad \frac{d}{dx}(\cos x) = -\sin x$$

The Chain Rule

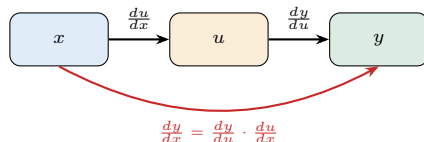
The Most Important Rule in All of Deep Learning

Statement

If y depends on u , and u depends on x (that is, $y = f(g(x))$):

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

Slopes of nested functions **multiply**.



Two Quick Applications

- $y = e^{\alpha x}$: let $u = \alpha x$, so $\frac{du}{dx} = \alpha$ and $\frac{dy}{dx} = e^u \cdot \alpha = \alpha e^{\alpha x}$.
- $y = \sin(\omega x)$: let $u = \omega x$, so $\frac{dy}{dx} = \cos u \cdot \omega = \omega \cos(\omega x)$.

Why This Is *The* Rule

A neural network is a long chain of nested functions. **Backpropagation** is nothing more than the chain rule applied repeatedly, multiplying slopes backwards from the loss to each weight.

Deep Dive: The Sigmoid Function

Setting Up the Derivation

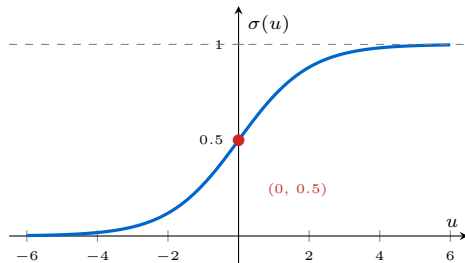
The Function

$$y = \sigma(u) = \frac{1}{1 + e^{-u}}, \quad u = \alpha x + \beta$$

- Squashes any real input into $(0, 1)$
- Output reads naturally as a **probability**
- The backbone of logistic regression

Shape Facts

- $\sigma(0) = 0.5$
- $\sigma(u) \rightarrow 1$ as $u \rightarrow +\infty$
- $\sigma(u) \rightarrow 0$ as $u \rightarrow -\infty$
- α controls how sharp the S-curve is



Deep Dive: Differentiating the Sigmoid

Chain Rule, Step by Step

Step 1 — Differentiate With Respect to u

Write $v = 1 + e^{-u}$, so that $y = v^{-1}$. Applying the chain rule:

$$\frac{dy}{du} = \frac{dy}{dv} \cdot \frac{dv}{du} = (-v^{-2}) \cdot (-e^{-u}) = \frac{e^{-u}}{(1 + e^{-u})^2}$$

Step 2 — Chain Through to x

Since $u = \alpha x + \beta$, we have $\frac{du}{dx} = \alpha$, giving

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = \alpha \cdot \frac{e^{-(\alpha x + \beta)}}{(1 + e^{-(\alpha x + \beta)})^2}$$

This is correct — but it is ugly, and it recomputes exponentials. There is a much better form.

The Elegant Form of the Sigmoid Derivative

Why Libraries Love It

Rewrite in Terms of σ Itself

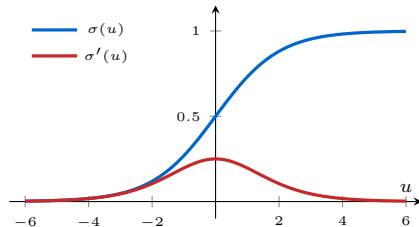
Split the fraction into two familiar pieces:

$$\frac{e^{-u}}{(1 + e^{-u})^2} = \underbrace{\frac{1}{1 + e^{-u}}}_{\sigma(u)} \cdot \underbrace{\frac{e^{-u}}{1 + e^{-u}}}_{1 - \sigma(u)}$$

Therefore:

$$\frac{d}{dx} [\sigma(\alpha x + \beta)] = \alpha \sigma(u) (1 - \sigma(u))$$

The forward pass already computed $\sigma(u)$; the derivative reuses it. **No new exponentials.**



Vanishing Gradients

σ' peaks at only 0.25 and is nearly 0 in both tails. Multiply many such factors through a deep chain and the gradient **dies** — the classic vanishing-gradient problem.

Higher-Order Derivatives

The Slope of the Slope

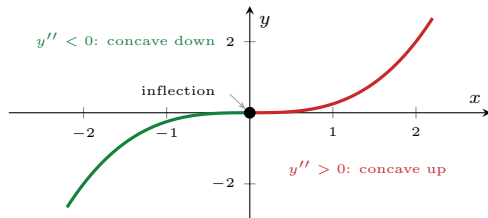
Definitions

- $y' = \frac{dy}{dx}$ — the slope
- $y'' = \frac{d^2y}{dx^2}$ — how the slope is changing
- $y''' = \frac{d^3y}{dx^3}$, and so on

The Intuition That Sticks

In a car: position $\rightarrow$ first derivative is **speed**
 $\rightarrow$ second derivative is **acceleration**.

On a graph y'' is **curvature**: positive holds water (bowl), negative spills it (dome).



One continuous curve. It bends one way before the inflection point and the other way after it.

Higher-Order Derivatives: Worked Example

Chain Rule Then Product Rule

Find y' and y'' for $y = \sin(x^2)$

① **First derivative** — chain rule with $u = x^2$, so $\frac{du}{dx} = 2x$:

$$y' = \cos(x^2) \cdot 2x = 2x \cos(x^2)$$

② **Second derivative** — now apply the product rule to $2x \cdot \cos(x^2)$:

$$\begin{aligned} y'' &= \frac{d}{dx}[2x] \cdot \cos(x^2) + 2x \cdot \frac{d}{dx}[\cos(x^2)] \\ &= 2 \cos(x^2) + 2x \cdot (-\sin(x^2) \cdot 2x) \\ &= 2 \cos(x^2) - 4x^2 \sin(x^2) \end{aligned}$$

Note the pattern: each differentiation of a composite pulls out another inner-derivative factor — why deep-network gradients accumulate many multiplied terms.

Optimization in One Variable

Finding the Bottom of the Valley

Step 1 — Where Is It Flat?

At a peak or a valley floor the tangent is horizontal:

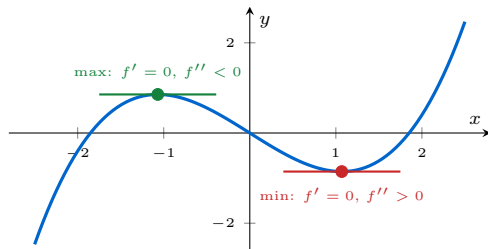
$$\frac{dy}{dx} = f'(x) = 0$$

Such points are called **critical** or **stationary** points.

Step 2 — Which Kind Is It?

Use the curvature:

- $\frac{d^2y}{dx^2} < 0 \Rightarrow$ concave down $\Rightarrow$ **maximum**
- $\frac{d^2y}{dx^2} > 0 \Rightarrow$ concave up $\Rightarrow$ **minimum**
- $\frac{d^2y}{dx^2} = 0 \Rightarrow$ inconclusive



Both points are flat — curvature is what separates them.

Optimization: Worked Example

A Complete Two-Step Check

Optimize $y = 4 - (x - 2)^2$

① Find critical points:

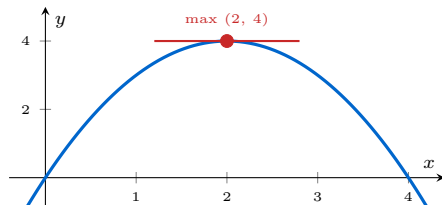
$$\frac{dy}{dx} = -2(x - 2) = 0 \implies x = 2$$

② Classify with curvature:

$$\frac{d^2y}{dx^2} = -2 < 0$$

Concave down, so this is a **maximum**.

③ Report the value: $y(2) = 4$.



Connecting to ML

Training minimises a loss. We look for $f' = 0$ with $f'' > 0$. In practice we do not solve $f' = 0$ algebraically — we **walk downhill** using the slope, which is gradient descent.

Section Take-Away: Differentiation

- ① **A derivative is the slope of the tangent** — “how steep is the curve right here”. Everything else is bookkeeping.
- ② It comes from a **secant line whose two points slide together** until the run shrinks to nothing.
- ③ The **chain rule multiplies slopes** through nested functions. Backpropagation is this rule, applied over and over.
- ④ The **second derivative is curvature**: bowl-shaped (positive) or dome-shaped (negative).
- ⑤ To optimize: find where the slope is **zero**, then use **curvature** to decide whether it is a peak or a valley.
- ⑥ $\sigma' = \sigma(1 - \sigma)$ peaks at just 0.25 — which is precisely why deep sigmoid networks suffer **vanishing gradients**.

Next: instead of the slope of a curve, the **area under** it $\Rightarrow$ **Integration**

Chapter 2 Quiz

Differentiation

- ❶ In one sentence: what is a derivative, geometrically?
- ❷ A curve is dome-shaped at a point. Is f'' positive or negative there?
- ❸ Why is $y = 2 + |x - 3|$ not differentiable at $x = 3$?
- ❹ What is the largest value σ' ever takes, and why does that cause vanishing gradients?
- ❺ At x_0 , $f'(x_0) = 0$ and $f''(x_0) > 0$. Maximum, minimum, or inconclusive?
- ❻ Differentiate $y = e^{3x}$ and $y = \sin(5x)$.
- ❼ Why is the chain rule called the foundation of backpropagation?

Answers are in the **Appendix** at the end of the deck.

Integration: The Area Under a Curve

The Core Idea, In One Picture

An Integral Is Just an Area

Differentiation asked: *how steep?*

Integration asks: *how much has accumulated?*

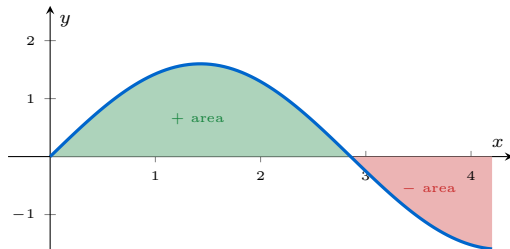
Definite Integral

$$\int_a^b f(x) dx$$

is the **signed area** trapped between the curve and the x -axis, from $x = a$ to $x = b$.

“Signed” Matters

Area **above** the axis counts as positive; area **below** counts as negative. They cancel.



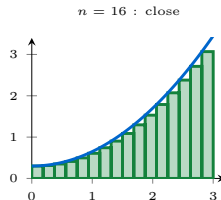
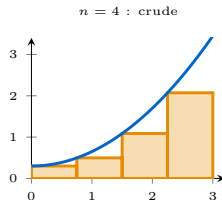
The integral returns the **net** of the green and red regions.

How the Area Is Actually Built

Riemann Sums: Slice, Sum, Shrink

The Recipe

- 1 Chop $[a, b]$ into n strips of width $\Delta x = \frac{b-a}{n}$.
- 2 Approximate each strip by a **rectangle**.
- 3 Add up all the rectangles.
- 4 Let $\Delta x \rightarrow 0$ ($n \rightarrow \infty$).



$$\int_a^b f(x)dx = \lim_{\Delta x \rightarrow 0} \sum_{k=0}^{n-1} f(a+k\Delta x) \Delta x$$

More strips $\Rightarrow$ the staircase hugs the curve $\Rightarrow$ the error vanishes in the limit.

The elongated “S” of $\int$ is literally an *S* for **sum**.

Indefinite Integration

Undoing a Derivative

The Anti-Derivative

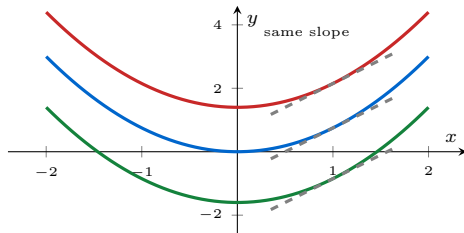
If $\frac{dy}{dx} = f(x)$, then

$$y = \int f(x) dx + C$$

Integration is differentiation run **backwards**.

Why the $+C$?

Differentiation **destroys constants**: x^2 , $x^2 + 5$ and $x^2 - 17$ all have slope $2x$. Running it backwards cannot tell you which you started from — a **many-to-one** map, the Section 1 invertibility issue returning.



A **family of curves**, vertically shifted. At any x every member has the identical slope.

The Fundamental Theorem of Calculus

The Bridge Between Slope and Area

Newton–Leibniz

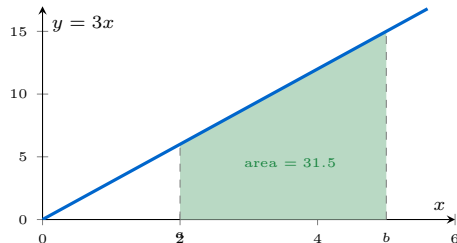
If F is any anti-derivative of f (that is, $\int f(x)dx = F(x) + C$), then

$$\int_a^b f(x) dx = [F(x)]_a^b = F(b) - F(a)$$

Why This Is Remarkable

Areas and slopes look unrelated, yet they are **inverse operations**. To get an area you never sum infinitely many rectangles — just find an anti-derivative and subtract.

$$\int_2^5 3x dx = \left[\frac{3}{2}x^2\right]_2^5 = \frac{3}{2}(25 - 4) = 31.5$$



Check by geometry: a trapezium of parallel sides 6 and 15, width 3 $\Rightarrow \frac{1}{2}(6 + 15)(3) = 31.5$. ✓

Standard Integration Rules

A Reference Table

Function $f(x)$	Integral $\int f(x) dx$	Example
$x^n \ (n \neq -1)$	$\frac{x^{n+1}}{n+1} + C$	$\int \sqrt{x} dx = \frac{2}{3}x^{3/2} + C$
$1/x$	$\ln x + C$	$\int_1^e \frac{1}{x} dx = \ln e - \ln 1 = 1$
e^{kx}	$\frac{1}{k}e^{kx} + C$	$\int e^{-5x} dx = -\frac{1}{5}e^{-5x} + C$
$\cos(kx)$	$\frac{1}{k} \sin(kx) + C$	$\int \cos(3x) dx = \frac{1}{3} \sin(3x) + C$

Table 1: Key rules of integration.

Sanity Check Habit

Always **differentiate your answer** to confirm you get the original integrand back. It takes five seconds and catches almost every algebra slip.

The Gaussian and Why Areas Matter in ML

Probability Density = Area

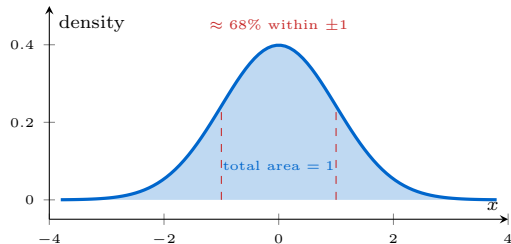
The Normalisation Condition

$$\int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = 1$$

The total area under a probability density must be exactly 1 — something must happen. The $\frac{1}{\sqrt{2\pi}}$ exists purely to force that.

Why the Exponent Must Be Negative

- With $-x^2/2$: the tails decay to 0, the area is finite, and we get the bell curve.
- With $+x^2/2$: the function explodes at both ends, the area is infinite, and no scaling can ever normalise it.



Reading a density: the height is not the probability — the **area** of a slice is. This is why integration underlies all of probability.

Section Take-Away: Integration

- ① **An integral is the area under a curve** — the accumulated total, as against the derivative's instantaneous steepness.
- ② That area is built by **slicing into rectangles and letting the slices become infinitely thin**. The $\int$ symbol is a stretched “S” for sum.
- ③ The **Fundamental Theorem** says differentiation and integration undo each other:
 $\int_a^b f = F(b) - F(a)$. Areas without ever summing rectangles.
- ④ The $+C$ appears because differentiation destroys constants — another many-to-one map that cannot be inverted.
- ⑤ In probability, **area is the probability**. A density's height means nothing on its own; the area of a slice is what you can bet on.

Next: real models have millions of parameters, not one $\Rightarrow$ **Multivariable Calculus**

Chapter 3 Quiz

Integration

- 1 What does $\int_a^b f(x) dx$ represent geometrically?
- 2 Suppose $\int_a^b f(x) dx = 0$ but f is not zero everywhere. What does that tell you?
- 3 Why does an indefinite integral carry a $+C$?
- 4 State the Fundamental Theorem of Calculus in words, without symbols.
- 5 Compute $\int_0^2 3x^2 dx$.
- 6 On a probability density plot, is the **height** the probability? If not, what is?
- 7 Why must the exponent in the Gaussian be negative?

Answers are in the **Appendix** at the end of the deck.

Multivariable Calculus & Optimization

From a Curve to a Landscape

Why One Variable Is Never Enough

The Jump

A real model has many parameters. Its loss is a function of all of them at once:

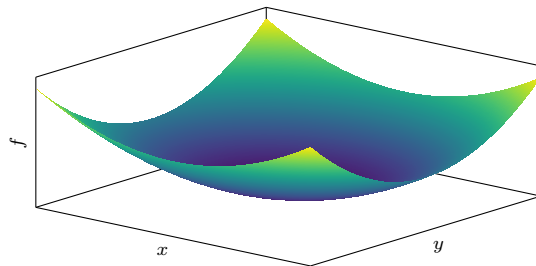
$$f(x_1, x_2, \dots, x_n) \longrightarrow \text{one number}$$

Many inputs in, a single scalar out.

The Mental Image

One variable $\rightarrow$ you walk along a **curve**.

Two variables $\rightarrow$ you stand on a **surface**, a hilly landscape. A million variables $\rightarrow$ same picture, just undrawable. **The 2-variable intuition carries over unchanged.**



A loss surface. Training means finding the lowest point on this landscape.

Partial Derivatives

Freeze Everything Else, Then Take a Normal Derivative

Definition

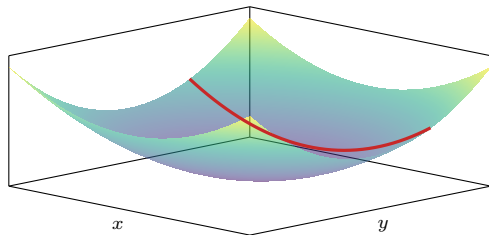
$$\frac{\partial f}{\partial x} = \lim_{\Delta x \rightarrow 0} \frac{f(x + \Delta x, y) - f(x, y)}{\Delta x}$$

Only x moves; y is held fixed.

The Picture

Slicing the surface with a vertical plane at $y = \text{constant}$ leaves an ordinary **curve**. The partial derivative is just the slope of that curve.

The curly ∂ is a reminder that other variables exist and are being deliberately held still.



The red curve is the surface cut at $y = 1$. Its slope is $\partial f / \partial x$.

Partial Derivatives: Basic Example

Treat the Other Variable as a Constant

Let $f(x, y) = x^2 \sin y$

With respect to x (so $\sin y$ is just a number sitting in front):

$$\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2) \cdot \sin y = 2x \sin y$$

With respect to y (so x^2 is just a number sitting in front):

$$\frac{\partial f}{\partial y} = x^2 \cdot \frac{\partial}{\partial y}(\sin y) = x^2 \cos y$$

The Only Skill Required

There is **no new calculus** here. Every rule from Section 2 — product, quotient, chain — applies unchanged. The single new habit is deciding which symbols are temporarily constants.

A Harder Partial Derivative

Product Rule Meets Chain Rule

The Function

$$g(x_1, x_2, x_3) = \underbrace{(x_1 + x_2 + x_3)^2}_{S^2} \cdot \underbrace{e^{-(x_1 + x_2^2 + x_3^3)}}_E$$

Write $S = x_1 + x_2 + x_3$. This is a **product**, so use the product rule; each factor then needs the **chain rule**.

Set-Up (do this once, reuse three times)

For S^2 , note $\frac{\partial S}{\partial x_i} = 1$ for every i , so

$$\frac{\partial(S^2)}{\partial x_i} = 2S$$

For E , only the exponent changes per variable:

$$\frac{\partial E}{\partial x_1} = -E, \quad \frac{\partial E}{\partial x_2} = -2x_2 E, \quad \frac{\partial E}{\partial x_3} = -3x_3^2 E$$

A Harder Partial Derivative: The Three Results

Assembling With the Product Rule

$$\text{Apply } \frac{\partial g}{\partial x_i} = \frac{\partial(S^2)}{\partial x_i} E + S^2 \frac{\partial E}{\partial x_i}$$

$$\frac{\partial g}{\partial x_1} = 2SE + S^2 E(-1) = SE [2 - S]$$

$$\frac{\partial g}{\partial x_2} = 2SE + S^2 E(-2x_2) = SE [2 - 2x_2 S]$$

$$\frac{\partial g}{\partial x_3} = 2SE + S^2 E(-3x_3^2) = SE [2 - 3x_3^2 S]$$

with $S = x_1 + x_2 + x_3$ and $E = e^{-(x_1 + x_2^2 + x_3^3)}$.

Notice: all three share the common factor SE . Spotting shared structure like this is exactly what makes automatic differentiation efficient — compute once, reuse everywhere.

The Gradient

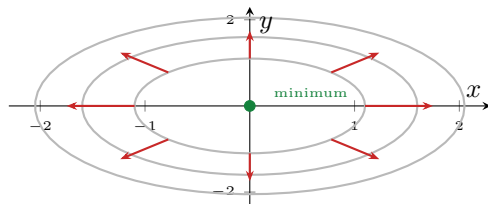
Collect All the Partial Derivatives Into One List

Definition

The **gradient** ∇f (“grad f ”) collects every partial derivative into one ordered list:

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \dots, \frac{\partial f}{\partial x_n} \right)$$

For $f(x, y) = x^2 \sin y$:
 $\nabla f = (2x \sin y, x^2 \cos y)$.



Grey rings = equal-height lines. Red arrows = gradient vectors.

A Gradient Is a **Vector**

And a Vector Is Just an Ordered List

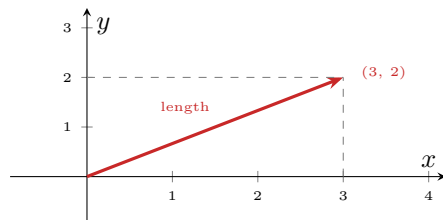
What “Vector” Actually Means

A **vector** is nothing more exotic than an **ordered list of numbers**. **Ordered** matters: slot 1 is the slope along x , slot 2 the slope along y — they are not interchangeable. One slot per parameter.

You have been writing vectors all along:
 $(2x \sin y, x^2 \cos y)$ is one.

Enough for Today

Drawn as an arrow, the list has a **direction** (which way to move) and a **length** (how strongly). That is all gradient descent needs.



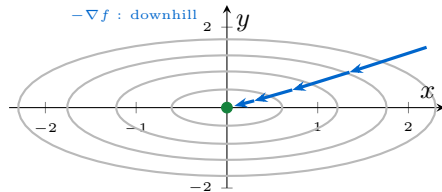
The list $(3, 2)$ drawn as an arrow. Same object, two ways of looking at it.

What the Gradient Means Geometrically

Three Properties, One Payoff

The Three Properties

- 1 ∇f points along **steepest ascent** — the fastest way uphill.
- 2 Its **length** is how steep that climb is: long arrow = steep, short = nearly flat.
- 3 It is always **perpendicular** to the contour through that point.



The Hiker's Analogy

On a foggy hillside you can only feel the ground underfoot. The gradient is the direction your toes say is **most steeply uphill**. To get **down**, walk the opposite way: $-\nabla f$.

Steps shorten near the bottom: the gradient shrinks there.

Gradient Descent

The Payoff: This Is How Models Train

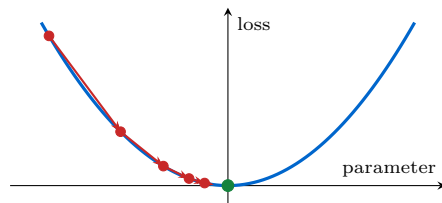
The Update Rule

$$x_i^{\text{new}} = x_i^{\text{old}} - \eta \frac{\partial f}{\partial x_i}$$

applied to every parameter, where η is the **learning rate** (the step size).

Reading the Rule in Plain English

- The **minus** sign means “go downhill”.
- Large partial derivative $\Rightarrow$ that parameter is strongly pushing the error up $\Rightarrow$ change it a lot.
- Flat ground $\Rightarrow$ derivative 0 $\Rightarrow$ the parameter stops. **You have arrived.**



Choosing η

Too small and training crawls. Too large and you overshoot the valley and bounce out. This is the single most-tuned knob in practical ML.

The Total Differential

Estimating a Change Before You Make It

A Warning About the Name

The **total differential** is *not* a “differential equation”. Despite the similar sound they are unrelated ideas: a differential equation is an equation you *solve* for an unknown function, whereas the total differential is just a **first-order estimate** of how much f will move. We will avoid the phrase “total differential equation” entirely.

The Formula

If each input shifts a little, the change in f is approximately the sum of each partial derivative times its own shift:

$$\Delta f \approx \frac{\partial f}{\partial x_1} \Delta x_1 + \frac{\partial f}{\partial x_2} \Delta x_2 + \dots + \frac{\partial f}{\partial x_n} \Delta x_n$$

Read it as: (*sensitivity to x_1*) $\times$ (*how much x_1 moved*), added up over all inputs.

Next class: this sum is exactly a **dot product**, and will be written compactly as $\nabla f \cdot \Delta x$.

Case Study: Software Company Cost

Putting Real Numbers In

The Setup

Operating cost depends on staff S , IT equipment E , and licenses L : $C = f(S, E, L)$.

Suppose the sensitivities are:

- $\frac{\partial C}{\partial S} = \$80,000$ per head, per year
- $\frac{\partial C}{\partial E} = \$2,000$ per unit
- $\frac{\partial C}{\partial L} = \$1,200$ per license

The Planned Change

Hire 10 staff, retire 2 equipment units, buy 3 licenses:

$$\Delta s = +10, \quad \Delta e = -2, \quad \Delta l = +3$$

The Estimate

$$\begin{aligned}\Delta C &\approx (80,000)(10) + (2,000)(-2) + (1,200)(3) \\ &= 800,000 - 4,000 + 3,600 \\ &= \boxed{\$799,600}\end{aligned}$$

Two Lessons

- 1 Staff **dominates** — the partial derivatives told us which lever matters; the rest is rounding error.
- 2 It is an **approximation**, valid for small changes. Hire 500 and you would need new offices, shifting the sensitivities themselves.

Optimization in Many Dimensions

Step 1: Find the Flat Spots

Stationary Condition

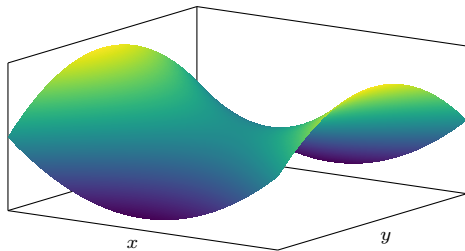
A landscape is flat at a peak, a valley floor, *or* a saddle. Flat means **every** partial derivative vanishes at once:

$$\frac{\partial f}{\partial x_1} = \frac{\partial f}{\partial x_2} = \dots = \frac{\partial f}{\partial x_n} = 0$$

written compactly as $\nabla f = 0$.

Flat Is Not Enough

In 1-D, flat meant peak or valley. In higher dimensions a third case appears: the **saddle point** — a minimum along one direction, a maximum along another.



A **saddle**: along one axis you sit at a valley floor; turn 90° and you are on a ridge top. Gradient zero, yet neither max nor min.

Step 2: Classify Using Curvature

The Table of Second Derivatives

The Hessian — a Table, For Now

In one variable, curvature was the single number f'' . In n variables we need every pairwise second derivative, arranged in a square table called the **Hessian**:

$$H = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix} \quad H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}$$

It is **symmetric**: $\frac{\partial^2 f}{\partial x \partial y} = \frac{\partial^2 f}{\partial y \partial x}$, so the table mirrors across its diagonal.

Reading Curvature Off the Hessian

The diagonal entries are the familiar “bowl or dome?” curvatures along each axis. The off-diagonal entries say how the axes **interact**. The next slide turns these into a verdict.

Classifying a Critical Point

Two Practical Tests

Test A — The Two-Variable Determinant Test

For $f(x, y)$, compute $D = f_{xx}f_{yy} - (f_{xy})^2$ at the critical point:

Condition	Meaning	Verdict
$D > 0$ and $f_{xx} > 0$	bowl in every direction	Local minimum
$D > 0$ and $f_{xx} < 0$	dome in every direction	Local maximum
$D < 0$	bends up one way, down another	Saddle point
$D = 0$	test gives no information	Inconclusive

Test B — The Separable Case (read it off by eye)

If the function has **no cross terms**, the Hessian is diagonal and you simply read the signs down the diagonal: all positive $\Rightarrow$ minimum, all negative $\Rightarrow$ maximum, mixed signs $\Rightarrow$ saddle.

Worked Example: A Three-Variable Minimum

Gradient Then Curvature

Optimize $f(x, y, z) = 3(x - 2)^2 + 4(y - 3)^2 + 5(z - 1)^2 + 12$

Step 1 — Find the flat spot. Set every partial derivative to zero:

$$\frac{\partial f}{\partial x} = 6(x - 2) = 0, \quad \frac{\partial f}{\partial y} = 8(y - 3) = 0, \quad \frac{\partial f}{\partial z} = 10(z - 1) = 0 \implies (x, y, z) = (2, 3, 1)$$

Step 2 — Check curvature. The second derivatives are constants, and there are no cross terms:

$$\frac{\partial^2 f}{\partial x^2} = 6, \quad \frac{\partial^2 f}{\partial y^2} = 8, \quad \frac{\partial^2 f}{\partial z^2} = 10, \quad \text{all mixed terms} = 0$$

Step 3 — Conclude (Test B)

The Hessian is diagonal with entries 6, 8, 10 — **all positive**, so the surface curves upward in every direction. The point $(2, 3, 1)$ is a **strict minimum**, with value $f(2, 3, 1) = \mathbf{12}$.

Section Take-Away: Multivariable Calculus

- 1 A **partial derivative** is an ordinary derivative with every other variable frozen. No new calculus — just a new habit.
- 2 The **gradient** is the list of all partial derivatives. It points **steepest uphill**, its length says how steep, and it cuts **across the contours**.
- 3 **Gradient descent** = repeatedly step in the direction $-\nabla f$. That single sentence is how essentially every model is trained.
- 4 The **total differential** estimates a change: sensitivity $\times$ movement, summed over inputs. It reveals **which lever matters**.
- 5 $\nabla f = 0$ is **not enough** in higher dimensions — flat could mean minimum, maximum, *or* saddle. Curvature settles it.
- 6 Curvature lives in the **Hessian**, a table of second derivatives. For now use the D -test or read a diagonal Hessian by eye.

Chapter 4 Quiz

Multivariable Calculus & Optimization

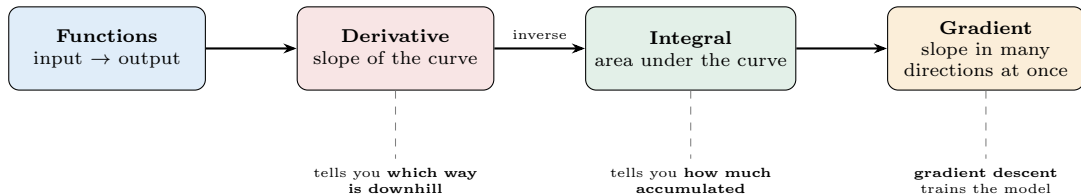
- 1 What does $\partial f / \partial x$ do with all the other variables?
- 2 At a point $\nabla f = \vec{0}$. Minimum, maximum, or not enough information?
- 3 Which direction decreases f fastest from where you stand?
- 4 In the software-cost case study, which lever dominated ΔC , and how could you tell?
- 5 For $f(x, y) = 3x^2 + 5y^2$: find the critical point, write the Hessian, and classify it.
- 6 What does a **long** gradient vector tell you, compared with a short one?
- 7 Why do gradient descent's steps naturally shrink as it nears a minimum?

Answers are in the **Appendix** at the end of the deck.

Wrapping Up

The Whole Deck on One Slide

How the Four Ideas Connect



The Sentence to Walk Away With

Training a model means using slopes to walk downhill on an error landscape you cannot see.

Where This Goes Next

Part II: Vectors & Matrices

You Already Have the Ideas

Everything ahead is the **same calculus** you just learned, written more compactly and applied to many parameters at once.

- The gradient you met today is a **vector** — next class gives vectors their full toolkit.
- Long sums like the total differential collapse into a single tidy operation.
- The Hessian becomes a **matrix** you can analyse directly.

Coming Up

- Vectors and the **dot product**
- Matrices as transformations
- **Eigenvalues** — the general curvature test
- Optimizing *with constraints*: “minimise cost, but stay under budget”

Same landscape, sharper tools for walking it.

Thank You

Questions?

Appendix: Quiz Answers

Answers — Chapter 1

Mappings, Functions & Their Shapes

- ❶ **No.** It is many-to-one: $f(-2) = f(2) = 4$, so the sign of x cannot be recovered. It fails the horizontal line test.
- ❷ $4y = -3x + 8 \Rightarrow y = -\frac{3}{4}x + 2$. Slope $m = -0.75$, intercept $c = 2$.
- ❸ Logs turn **products into sums**. A product of many small probabilities collapses below the smallest representable number, but its logarithm is a moderate number that adds up safely.
- ❹ **Many-to-one.** The magnitudes of the negative inputs are destroyed — and with them the gradient signal for those inputs.
- ❺ At most **4** turning points (degree n gives at most $n - 1$). More degree means more wiggle, and so more capacity to **overfit**.
- ❻ **True.**

Answers — Chapter 2

Differentiation

- ❶ The **slope of the tangent line** to the curve at that point — how steep it is right there.
- ❷ **Negative.** A dome is concave down.
- ❸ Approaching from the left gives slope -1 ; from the right, $+1$. They disagree, so no single tangent slope exists at the corner.
- ❹ σ' peaks at **0.25** (at $u = 0$) and is near 0 in both tails. Multiplying many such factors down a deep chain drives the gradient toward zero.
- ❺ A **minimum** — flat, and curving upward.
- ❻ $\frac{dy}{dx} = 3e^{3x}$ and $\frac{dy}{dx} = 5 \cos(5x)$.
- ❼ A network is a long chain of nested functions. Backpropagation is the chain rule applied repeatedly, multiplying slopes backwards from the loss to each weight.

Answers — Chapter 3

Integration

- ❶ The **signed area** between the curve and the x -axis, from a to b .
- ❷ The area **above** the axis exactly cancels the area **below** it.
- ❸ Differentiation **destroys constants** (x^2 , $x^2 + 5$ and $x^2 - 17$ all have slope $2x$), so running it backwards cannot recover which constant you started from.
- ❹ Differentiation and integration **undo each other**: to find an area, find an anti-derivative and subtract its values at the two endpoints.
- ❺ $[x^3]_0^2 = 8 - 0 = 8$.
- ❻ **No**. The **area** of a slice is the probability; the height alone means nothing on its own.
- ❼ With a positive exponent the function explodes at both ends, the total area is infinite, and no scaling could ever normalise it to 1.

Answers — Chapter 4

Multivariable Calculus & Optimization

- ❶ It **holds them all constant**, so only x is allowed to move.
- ❷ **Not enough information.** It could be a minimum, a maximum, or a saddle point. Curvature settles it.
- ❸ Along $-\nabla f$, the direction of steepest descent.
- ❹ **Staff.** Its term was $(80,000)(10) = 800,000$, against $-4,000$ and $3,600$ — the partial derivatives showed which lever mattered.
- ❺ $\nabla f = [6x, 10y]^T = \vec{0} \Rightarrow (0, 0)$. $H = \begin{bmatrix} 6 & 0 \\ 0 & 10 \end{bmatrix}$, diagonal with both entries positive $\Rightarrow$ a **strict minimum**.
- ❻ A long gradient means the ground is **steep**; a short one means it is nearly flat.
- ❼ The gradient itself shrinks near a minimum, and the step is proportional to the gradient — so the steps shrink automatically.